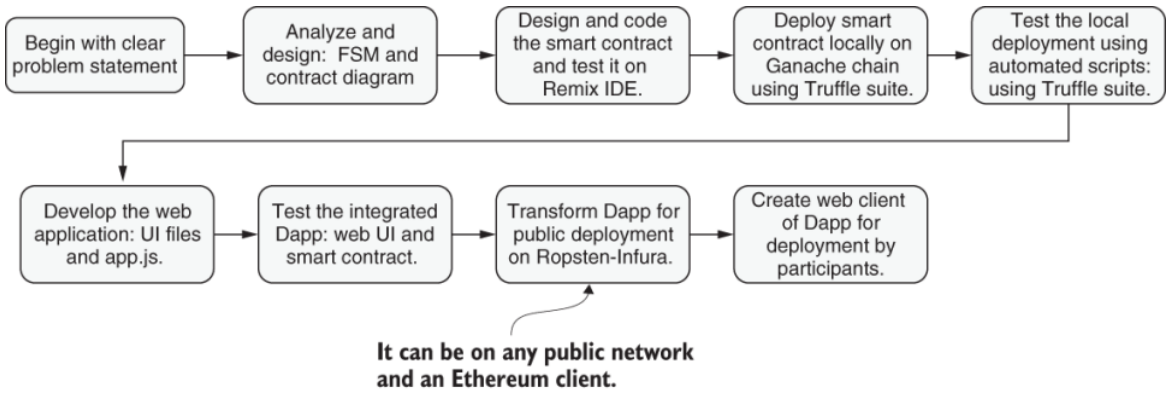


Figure 11.1 A roadmap to developing Dapps for Ethereum blockchain

Figure 11.1 shows the roadmap from local deployment on the Ganache test chain to the public implementation involving the public network Ropsten with Infura as web3 provider. Although Ropsten is the network used in this chapter, you can choose to work another public Ethereum network, including mainnet, simply by configuring your deployment. The roadmap will help you can navigate this chapter as well as your future Dapp development.

11.3 Problem description

Let’s begin by articulating a clear statement for the problem to be solved.

> **PROBLEM STATEMENT** An undergraduate data-intensive computing certificate (hereafter referred to as DCC) program requires an enrollee to complete four categories of courses and an average grade-point average (GPA) of at least 2.5 in these courses. Although the details of this program are stated in an undergraduate catalog page, there is no traditional tool implemented for verifying the fulfillment of a certificate. The objective of this project is to build an *independent* blockchain-based tool so that any student interested in this certificate program can use this tool. A student enrolled in the certificate program can self-check their progress through the program from anywhere.